DSA MCQ Quiz

Data Structures & Algorithms | 10 questions | 1 mark each | Total: 10 marks | Choose the one correct option. Answer key is on the last page.

Q1. What is the time complexity of binary search on a sorted array of n elements? [1 Mark]

(A) O(n)

(B) O(log n)

(C) O(n log n)

(D) O(1)

Q2. Which data structure follows the LIFO (Last In, First Out) principle? [1 Mark]

(A) Queue

(B) Stack

(C) Linked List

(D) Heap

Q3. What is the worst-case time complexity of Quick Sort? [1 Mark]

(A) O(n log n)

(B) O(n)

(C) $O(n^2)$

(D) O(log n)

Q4. Which data structure is used to implement Breadth-First Search (BFS)? [1 Mark]

(A) Stack

(B) Priority Queue

(C) Queue

(D) Hash Map

Q5. What is the average-case time complexity of searching in a hash table with a good hash function? [1 Mark]

(A) O(1)

(B) O(log n)

(C) O(n)

(D) $O(n^2)$

Q6. In-order traversal of a Binary Search Tree (BST) produces nodes in which order? [1 Mark]

(A) Random order

(B) Descending order

(C) Level order

(D) Ascending (sorted) order

Q7. Which algorithm finds the shortest path from a source in a weighted graph with non-negative edge weights? [1 Mark]

(A) Dijkstra's Algorithm

(B) Kruskal's Algorithm

(C) DFS

(D) Prim's Algorithm

Q8. What is the height of a complete binary tree with n nodes? **[1 Mark]**

(A) O(n)

(B) O(n log n)

(C) O(log n)

(D) O(1)

Q9. Which of the following sorting algorithms is stable and has O(n log n) worst-case time? **[1 Mark]**

(A) Quick Sort

(B) Heap Sort

(C) Selection Sort

(D) Merge Sort

Q10. What is the minimum number of edges required for a connected undirected graph with n vertices? **[1 Mark]**

(A) n

(B) n - 1

(C) n + 1

(D) n / 2

Answer Key

Q	Topic	Correct Answer	Why	Marks
1	Searching	**(B)** O(log n)	Each step halves the search space, so at most log n comparisons are needed.	1
2	Stack	**(B)** Stack	A stack removes the most recently inserted element first.	1
3	Sorting	**(C)** $O(n^2)$	A consistently bad pivot (e.g. already sorted input with first/last pivot) gives $O(n^2)$.	1
4	Graphs (BFS)	**(C)** Queue	BFS explores level by level, which needs FIFO order, so a queue.	1
5	Hashing	**(A)** O(1)	A good hash function spreads keys evenly, so lookups take constant time on average.	1
6	Trees (BST)	**(D)** Ascending (sorted) order	Left, Root, Right on a BST visits keys in increasing order.	1
7	Graphs (Shortest Path)	**(A)** Dijkstra's Algorithm	Dijkstra greedily picks the closest unvisited vertex; Kruskal and Prim build spanning trees.	1
8	Trees (Heap)	**(C)** O(log n)	Each level doubles the node count, so the height is floor($\log_2 n$).	1
9	Sorting	**(D)** Merge Sort	Merge Sort preserves the order of equal elements and always runs in O(n log n).	1
10	Graphs (Basics)	**(B)** n - 1	A spanning tree connects n vertices using exactly n - 1 edges.	1

Total Marks: 10